Filtered Attention for Long-Context Retrieval

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Abstract

Long-context language models often fail to retrieve facts buried in the middle of their input. We introduce filtered attention, a lightweight scoring layer that prunes irrelevant key blocks before the softmax. Code is available at github.com/ada/filtered-attention.

1 Introduction

Retrieval failures are not uniform across the context window. Facts near the beginning and the end of the input are recovered reliably, while facts in the middle are frequently ignored, a pattern earlier work calls the lost-in-the-middle effect.

We argue that much of this failure comes from attention mass being spread over many irrelevant blocks. When most of the probability mass goes to filler text, the few tokens that matter receive too little weight to influence the output.

Our approach keeps the base model frozen. We add a small scoring layer that looks at each block of keys and decides whether the block is worth attending to at all, which keeps training cheap and the method easy to retrofit.

arXiv:2401.00001v1 [cs.CL] 2 Jan 2024

2 Method

Filtered attention improves retrieval recall by 11.4 points at 128k tokens on NeedleBench, as shown in Table 1.

Because pruned blocks never enter the softmax, the savings grow with context length, which makes the method most useful exactly where long-context models are most expensive.

3 Related work

Sparse attention has a long history, and retrieval heads were recently identified in long-context models.

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